

Homework 3

STATISTICAL MODELING AND LEARNING

due: **Friday, February XX, 2026 at 10:00a**

Instructions: Please submit solutions on Populi as a knitted markdown pdf. For handwritten work, please scan your solution pages and include in the final pdf. Homeworks can be completed individually or in groups of up to three people. Names of all group members must be included on the writeups.

Problem 1: Time Series Decomposition. Download the file `airline_passengers.csv` from the course website. This dataset contains monthly totals of international airline passengers (in thousands) from 1949 to 1960.

- Plot the time series. What components do you observe (trend, seasonality, cyclical)?
- Fit an additive decomposition model using linear regression, with a linear trend and monthly dummy variables for seasonality. Report the fitted model and interpret the coefficient on the trend term and at least two of the seasonal coefficients.
- Plot the fitted values from part (b) overlaid on the original data. Then plot the residuals over time. Do the residuals look random, or is there structure remaining?
- The variance of the data appears to increase over time. Fit a multiplicative model by taking the log of passengers and regressing on the same predictors (linear trend and monthly dummies). Compare the residual plots from the additive and multiplicative models. Which model appears more appropriate?
- Using your preferred model, what is the predicted number of passengers for June 1961? How confident are you in this prediction?

Problem 2: Autoregressive Models. Download the file `daily_temperature.csv` from the course website. This dataset contains daily average temperatures (in Fahrenheit) for a city over one year.

- Plot the time series. Does there appear to be a clear trend? Seasonality?
- Create a new variable `lag1` that contains the previous day's temperature. Plot today's temperature against yesterday's temperature. What do you observe?
- Fit an AR(1) model: regress today's temperature on yesterday's temperature. Report the fitted model and interpret both coefficients.
- Examine the ACF (autocorrelation function) of the residuals from your AR(1) model. Are there any significant autocorrelations remaining? What does this tell you about model adequacy?
- Based on your AR(1) model, if today's temperature is 72°F, what is the predicted temperature for tomorrow? What about two days from now?
- Is this time series a random walk? Construct a 95% confidence interval for β_1 (the coefficient on lagged temperature) and use it to assess whether $\beta_1 = 1$ is plausible.

Problem 3: Filtering Noisy Data. This problem introduces the idea of filtering—estimating a true underlying signal from noisy observations.

Suppose we are tracking the true price of an asset, but our observations are noisy. Let θ_t denote the true price at time t , and Y_t denote the observed (noisy) price. Assume:

- Observation: $Y_t = \theta_t + v_t$, where $v_t \sim N(0, V)$ is measurement noise
 - State: $\theta_t = \theta_{t-1} + w_t$, where $w_t \sim N(0, W)$ is the random change in true price
- a. In words, explain what this model assumes about how the true price evolves over time. What type of time series model is this for θ_t ?
 - b. Download the file `noisy_prices.csv` from the course website. Plot the observed prices Y_t over time.
 - c. One simple approach to estimate the true price is a moving average. Compute a 5-day centered moving average of the observed prices and plot it alongside the original data. What are the advantages and disadvantages of this approach?
 - d. The Kalman filter provides an optimal way to estimate θ_t . At each time t , we:
 - (i) **Predict:** Our forecast is $a_t = m_{t-1}$ (yesterday’s estimate), with uncertainty $R_t = C_{t-1} + W$
 - (ii) **Update:** Combine prediction with observation using Kalman gain $K_t = R_t / (R_t + V)$:

$$m_t = a_t + K_t(Y_t - a_t)$$

Explain in words what the Kalman gain K_t represents. When is K_t close to 1? When is it close to 0?

- e. Using $V = 4$, $W = 0.5$, and starting values $m_0 = Y_1$ and $C_0 = 10$, implement the Kalman filter in R and plot the filtered estimates alongside the observed data. How does it compare to the moving average?
- f. Experiment with different values of W (try $W = 0.05$ and $W = 5$). How does changing W affect the filtered estimates? Relate this to the Kalman gain formula.

Problem 4: Logistic Regression and Bank Marketing. Download `bank-full.csv` and `bank-names.txt` from the course website. The data come from a Portuguese bank’s direct marketing campaigns (phone calls) to promote a term deposit. The outcome of interest is whether the client subscribed to the term deposit (y : “yes” or “no”).

Key variables (see `bank-names.txt` for full descriptions):

- **age:** client age (numeric)
- **balance:** average yearly balance in euros (numeric)
- **previous:** number of contacts performed before this campaign for this client (numeric)
- **poutcome:** outcome of the *previous* marketing campaign (categorical: “unknown”, “other”, “failure”, “success”)

- **job**: type of job (categorical: e.g., “management”, “blue-collar”, “retired”, “student”, etc.)

Use only these variables (and **y**) in the problem below.

- What fraction of clients in the dataset subscribed to the term deposit? Create an exploratory plot showing how subscription rate varies by one categorical predictor (e.g., **job** or **poutcome**). What do you observe?
- Fit a logistic regression model predicting subscription (**y**) from **balance**, **age**, and **previous**. Write out the fitted model. Interpret the coefficient on **balance** in terms of odds: for every 1000 euro increase in yearly balance, how do the odds of subscribing change?
- In the model from part (b), interpret the coefficient on **previous**. Does having been contacted in a previous campaign appear to be associated with higher or lower probability of subscribing in this campaign?
- Add **poutcome** (treat as categorical; choose a reference category) to the model. Is previous campaign success (**poutcome** = “**success**”) associated with higher probability of subscribing? Interpret the estimated coefficient for previous success in terms of odds.
- Using your model from part (d), what is the predicted probability of subscribing for a client with **balance** = 2000 euros, **age** = 45, **previous** = 2, and **poutcome** = “**success**”? How does this compare to a client with the same **balance**, **age**, and **previous** but **poutcome** = “**failure**”?
- In one or two sentences, how might the bank use a model like this to decide whom to contact in a future campaign?